



Amit Jagannath Patil, PhD

Postdoctoral Research Fellow

Institute of Urban Environment, Chinese Academy of Sciences, China

Email: amitpatil12.89@gmail.com · Phone: +91 9987171016

Date of birth: 12/07/1989 · Gender: Male

Nationality: India · Hometown: Mumbai

ORCID: [0000-0001-8958-3895](https://orcid.org/0000-0001-8958-3895) · ResearchGate : [Amit-Patil-13](https://www.researchgate.net/profile/Amit-Patil-13)
[Google Scholar](https://scholar.google.com/citations?user=amitpatil12.89) · [LinkedIn](https://www.linkedin.com/in/amitpatil1289) · Research Portfolio: aquadatapod.github.io

Core competencies: Biodiversity & Community Ecology · Advanced Quantitative Modeling · Large-scale Ecological Data Synthesis; Field & Experimental Methodologies · Scientific Writing & Mentoring; Research communication

RESEARCH PROFILE

Dr. Amit Jagannath Patil is a Postdoctoral Research Fellow at the Institute of Urban Environment (IUE), Chinese Academy of Sciences (CAS) in Xiamen, China. His research lies at the intersection of macroecology, watershed biodiversity, and pollution ecology, with a primary focus on understanding how multi-scale environmental drivers shape aquatic and semi-aquatic communities.

Specializing in advanced quantitative data synthesis and ecological modeling in R, his work extensively explores river watershed ecosystems to decouple the complex mechanisms driving macroinvertebrate biodiversity across **taxonomic, functional, and phylogenetic** dimensions. While his foundational doctoral research established structural equation frameworks (piecewiseSEM) to evaluate macrobenthic community functioning across riverine mesohabitats, his current postdoctoral work scales these ecological principles globally. By integrating multi-taxa datasets, he models how emerging anthropogenic stressors and global chemical gradients alter trait dimensionality, biodiversity patterns, and ecosystem resilience. Alongside his research, he is deeply committed to academic mentorship, guiding postgraduate scholars, and delivering specialized training in multivariate ecological data analysis and scientific communication.

EDUCATION

- **2017–2023** — PhD in Environmental Science, Chang'an University, Xi'an, China
- **2010–2012** — MSc (Hons) Zoology (Oceanography), Mumbai University, Mumbai, India
- **2007–2010** — BSc Zoology, Mumbai University, Mumbai, India

EMPLOYMENT / RESEARCH POSITIONS

- **2023–present** — *Postdoctoral Research Fellow (PDF)*
Institute of Urban Environment, Chinese Academy of Sciences, Xiamen, China
 - Investigate the global distribution, accumulation scales, and ecological risks of emerging pollutants (microplastics, pesticides, pharmaceuticals) within aquatic and terrestrial wildlife communities.

- Apply advanced data-driven approaches, meta-analyses, and machine-learning frameworks to evaluate trait-mediated toxicity and community-level vulnerability.
- Synthesize large-scale global datasets through extensive literature mining and advanced statistical modeling in R.
- **2015–2017 — Project Assistant II / Research Assistant**
CSIR–National Institute of Oceanography, Goa, India
 - Studied benthic meiofauna from the Central Indian Ocean Basin; recorded coral distribution along the west coast of India; SCUBA diving on scientific expeditions.
- **2014–2015 — Junior Research Fellow (JRF)**
Salim Ali Centre for Ornithology and Natural History, South India Centre of Wildlife Institute of India, Ministry of Forest and Climate Change, Coimbatore, India
 - Behavioral study and identification of coastal and pelagic birds; studied water chemistry and habitat structure of shorebird sites; collected distributional data of coastal birds.
- **2012–2014 — Project Assistant II / Research Assistant**
CSIR–National Institute of Oceanography, Mumbai, India
 - Collected meso-zooplankton from estuarine and marine environments; analyzed water physico-chemical variables; biologist aboard RV Sindhu Sadhana (cruise SSD Tr-08).

PUBLICATIONS

Peer-reviewed articles

1. Iheanacho CC, Zhang D, Xu YY, **Patil AJ**, Ayiera NK, Sethar SA (2026). Gradient-boosting threshold identification for developmental toxicity: A BPAF case-study framework for early-stage risk assessment. *Environmental Pollution*. <https://doi.org/10.1016/j.envpol.2026.128645> (SCI; 2025, IF = 7.2, Ranking = 52/395, ENVIRONMENTAL SCIENCES)
2. Melnikova A, Guo ZF, Woolway RI, Ao SC, **Patil AJ**, Xu YY, Ayiera KN (2026). Multiscale contrasts in occurrences and patterns of pesticide contamination between river water and groundwater. *ACS ES&T Water*, acsestwater.5c01459. <https://doi.org/10.1021/acsestwater.5c01459> (SCI; 2025, IF = 4.5, Ranking = 147/395, ENVIRONMENTAL SCIENCES)
3. Gavas S, **Patil AJ** (corresponding author) *, Gurav M (2026). Global research trends on heavy metal contamination in bivalve. *Regional Studies in Marine Science*, 96, 104886. <https://doi.org/10.1016/j.rsma.2026.104886> (SCI; 2025, IF = 2.7, Ranking = 70/205, ECOLOGY)
4. Ao SC, Cower W, Resh VH, Chiu MC, Zhang D, Xu YY, **Patil AJ**, Guo ZF, Sun X, Xu YY (2026). Biogeographic biases in microplastic ingestion research mask biodiversity and consumption risks across benthic invertebrates. *Water Research*, 291, 125115. <https://doi.org/10.1016/j.watres.2025.125115> (SCI; 2025, IF = 12.8, Ranking = 15/395, ENVIRONMENTAL SCIENCES)
5. Wang Z, Li H, **Patil AJ** (2019). China's 'Silicon Valley' must not trash its environment. *Nature*, 574(7779):486. [doi:10.1038/d41586-019-03207-x](https://doi.org/10.1038/d41586-019-03207-x) (SCI; 2025, IF = 56.1, Ranking = 1/140, MULTIDISCIPLINARY SCIENCES)
6. Sivadas SK, **Patil AJ** (2023). Moving beyond traditional macrofaunal community structure studies in the Indian Ocean continental shelf: a research synthesis based on research weaving. *Environmental Science and Pollution Research*. [doi:10.1007/s11356-022-25030-0](https://doi.org/10.1007/s11356-022-25030-0) (SCI; 2025, Ranking = 86/359, ENVIRONMENTAL SCIENCES)

7. **Patil AJ**, Wang Z, He X, Li P, Yan T, Li H (2023). Understanding the effect of environment on macrobenthic invertebrate in naturally occurring repeated mesohabitats from the warm-temperate zone river. *Ecohydrology & Hydrobiology*. doi:10.1016/j.ecohyd.2022.10.001 (SCI; 2025, IF = 2.8, Ranking = 84/205, ECOLOGY)
8. Wang Z, Chen C, Wu Q, Wan L, **Patil AJ**, He X (2019). Microbial abundance and enzyme activity in sediments of the rapid-pool-benchland systems in the natural Duliu River of China. *Journal of Earth Sciences & Environmental Studies*, 4(6):749–762. doi:10.25177/JESES.4.6.RA.602 (Non-SCI Journal)
9. **Patil AJ**, Babu G, Shirke B, Rao SB, Manchi S, Quadros G (2019). Record of a few pelagic bird sightings along Sindhudurg coast, Maharashtra. *Ambient Science*, 6(2). PDF (ESCI; 2025, JCI = 0.02, Ranking = 188/189, ECOLOGY)
10. **Patil AJ**, Shirke B, Babu S, Rao GB, Quadros G (2016). Unusual weather condition causing the transfer of seahorses *Hippocampus kuda* onto the sandy beach of Sindhudurg district, Maharashtra. *Current Science*, 111(5):792–793. PDF (SCI; 2025, IF = 0.9, Ranking = 94/140, MULTIDISCIPLINARY SCIENCES)
11. Ao SC, Liu XY, **Patil AJ**, Chiu MC, Cai QH (2026). Trait dimensionality and coding shape estimates of functional diversity. *Journal of Watershed Ecology*, 1(2):10012-10012 <https://doi.org/10.70322/jwe.2026.10012> (Non-SCI Journal)

Under review / in preparation

1. **Patil AJ**, Stubbins A, Ao SC, Chiu MC, Zhang D, Guo ZF, Xu YY. Small birds in terrestrial habitats face highest microplastic exposure risk. *Global Environmental Change* (under review).
2. **Patil AJ**, Malcorps W, Prata J, Azevedo RMS, Das K, Newton R, Zhang D, Ao SC, Xu YY. Global patterns for size-dependent extreme microplastic contamination in shrimp and its implications for food safety. *Global Ecology and Biogeography* (under review).
3. Chiu M-C, **Patil AJ**, Melnikova A, Warrier A, Ibrahim M, Frank Y, Kalangutkar N, Nasyrova E, Ao S-C, Resh V, Xu Y-Y. Autonomous Active-Learning Can Accelerate Synthesis of Environmental Information for Emerging Contaminants. (under review).
4. **Patil AJ**, Melnikova A, Xu YY. Ecological performance ceiling framework for neonicotinoid impacts on birds. (in preparation).
5. **Patil AJ**, Melnikova A, Lamichhane P, Chinedu I, Xu YY. Sublethal and Population-Level Effects of Neonicotinoid in Songbirds. (in preparation).
6. Melnikova A, Ao SC, Guo ZF, Bank M, Xu YY, **Patil AJ**, Ayiera K. Hierarchical distribution and patterns of psychoactive substances in global rivers. *Environmental Science & Technology* (under review).
7. Ayiera KN, Nava V, Leoni B, Guo ZF, **Patil AJ**, Melnikova A, Xu YY (2025). Occurrence and ecological risks of microplastic pollution in global lakes. *Journal of Hydrology*. (under review).
8. Yan F, He X, Li H, Li P, **Patil AJ**, Peng J, Qian H, Wang Z (2025). Microbial diversity of sediment in the rapid–pool–benchland systems of four rivers, Xi'an, and strategies for restoration of degraded rivers. *Water Biology*. (under review).
9. Xie ZX, **Patil AJ**, Xu YX, Wang MY, He YQ, Chen C, Zhang D. (2026). Heavy metals as drivers of life-history convergent evolution in anurans. *Proceedings of the Royal Society B* (under review).

Conference proceedings

- **Patil AJ**, Wang Z, Le H, Pangen L, He L, Ting Y (2019). Benthic chlorophyll and environmental variables along different macro systems of the rivers, Shaanxi, China. *The Second River Basin Ecology Forum*, Nanchang, China.
- **Patil AJ**, Rakesh PS, Sukumaran S (2019). Mesozooplankton and its distribution in stressed tropical estuary. *The Second International Symposium on Water Resource Management and Comprehensive Reclamation of River Basin*, Xi'an, China.
- **Patil AJ**, Patil V, Mote S, Nagesh R, Thomas L, Ingole B (2015). Waterbirds as an ecological tool for assessing the environmental status. *International Conference on Climate Change and Sustainability*, Thakur college of Science and Commerce, Mumbai, India.

FUNDING, AWARDS & HONOURS

- **2025** — Second Prize (~USD 84,500), SUSONG Cup, Future Information Category, China
- **2022** — Excellent Innovation Achievement Award at 6th Graduate Student Innovation Exhibition, Shaanxi Provincial Department of Education, China
- **2020** — Chinese Government Outstanding International Student Award, CSC, China
- **2020** — Academic Star Award, International Education School, Chang'an University, Xi'an, China
- **2019** — Best Paper Presenter, 2nd River Basin Ecology Forum, Nanchang, China
- **2019** — Excellent International Student, Chang'an University, Xi'an, China
- **2019** — Academic Star Award, International Education School, Chang'an University, Xi'an, China
- **2018** — Excellent Ambassador, International Education School, Chang'an University, Xi'an, China
- **2017** — MHRD (India)–CSC (China) Bilateral Scholarship Award

TEACHING, MENTORING & INVITED LECTURES

- **2023–present** — *Multivariate ecological data analysis and scientific writing and Mentoring of PhD scholars*; training delivered in , IUE, CAS
- **2019** — Research Journey — Past and Present, IUE, CAS, China
- **2019** — *Biological Diversity*, Professor Assistantship, Chang'an University, Xi'an, China
- **2018** — *Marine Biodiversity & Ecology*, Professor Assistantship, Chang'an University, Xi'an, China
- **2020** — *Marine and Aquatic Research — Challenges*, Expert Invitation, OikoEssence, Mumbai, India
- **2019** — *Career Opportunity Abroad*, Resource Person, R.J. College, Mumbai, India
- **2019** — *Statistical Techniques in Ecology*, Resource Person, M.D. College, Mumbai, India
- **2019** *Wetland Research*, Resource Person, M.D. College of Arts, Science and Commerce, India
- **2019** — *Wetland Research*, Resource Person, M.D. College of Arts, Science and Commerce, India
- **2019** — *Avenues and Scholarships for Higher Education in Zoology*, Resource Person, D.G. Ruparel College, Mumbai, India
- **2019** — Zoology Alumni Achiever's Talk, Sathaye College, Mumbai, India
- **2019** — Delegate, International Conference on Ecology, Ethology and Environment Management, B.N. Bandodkar College, Mumbai, India

RESEARCH PROJECTS AS PI OR MAJOR PARTICIPANT

- **2019** — Central University basic scientific Research business fee subsidy project, Leading talent project, 300102299303, Research on microorganisms and aquatic insects in natural river rapids-deep pools-river beaches, 200,000 yuan, completed. **(participation)**

PROFESSIONAL STATISTICAL & METHODOLOGICAL TRAINING

- **2021** — Structural Equation Modeling Using R, PR Statistics Ltd, Glasgow, United Kingdom
- **2020** — Hierarchical Modelling of Species Communities, University of Helsinki, Helsinki, Finland
- **2020** — Introduction to Generalized Linear Models Using R, PR Statistics Ltd, Glasgow, UK
- **2020** — Environmental Analytics and Data Modelling, The Environmental Analytics, Pune, India
- **2020** — Statistics and R Programming, The Environmental Analytics, India
- **2020** — R for Ecologists and Evolutionary Biologists, PR Statistics Ltd, Glasgow, United Kingdom
- **2019** — GLM, Nonlinear and GAM Models, PR Statistics Ltd, Glasgow, United Kingdom

ACADEMIC SERVICE

Peer Reviewer for International Journals — Aquatic Sciences · Discover Sustainability · Thalassas

ACADEMIC VISITS & INTERNATIONAL ENGAGEMENT

- **October 2024** — *COP-16 Biodiversity Convention, Cali, Colombia*
Member of the Chinese delegation, “Peace with Nature” theme, China Pavilion **(delegate)**
- **October 2023** — *International Workshop Regreen: Nature Based Solutions, Ningbo, China* **(delegate)**

CERTIFICATIONS

- Certified Open Water SCUBA Diver, Scuba Schools International (SSI)
- GIS and Remote Sensing, University of Mumbai
- Certificate Course in Nature Interpretation (CCNI), Ecofolk’s Pvt.Ltd., Mumbai

PROFESSIONAL MEMBERSHIPS

- Life Member, Ocean Society of India (2014–present)

LANGUAGES

Mother tongue — Marathi

Other language(s)	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
Hindi	C2	C2	C2	C2	C1
English	C2	C2	C2	C2	C1
Chinese	A2	A1	A2	A2	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user